

Name: _____

Quiz 5 (September 30, 2022)

Multiple choice questions (1-5): select one answer from the five options.

Fill in the blanks (6,7): write your answer in the space given within the question.

Short answer (8-10): write your answer in the space given below the question.

2 points will be awarded for each correct answer (total 20)

1. A volume of air occupying 12.0 dm^3 at 98.9 kPa is compressed to a pressure of 119.0 kPa . The temperature remains constant. What is the new volume?
 - a. 99.7 dm^3
 - b. 8.65 dm^3
 - c. 9.97 dm^3
 - d. 10 dm^3
 - e. 8.89 dm^3
2. If you expect a chemical reaction to produce 4.38 dm^3 of oxygen, O_2 , at 19°C and 101 kPa , what will be the volume at 25°C and 101 kPa ?
 - a. 4.47 dm^3
 - b. 5.38 dm^3
 - c. 4.41 dm^3
 - d. 3.38 dm^3
 - e. 44.7 dm^3
3. Which statement is wrong?
 - a. In Boyle's law, only temperature needs to be constant
 - b. Both pressure and amount of gas are constant in Charles's law
 - c. At a fixed temperature for a given amount of gas, the product of pressure and volume will be constant
 - d. At STP, molar gas volume is found to be 22.4 dm^3
 - e. From ideal gas law, $P/n = (R/V) \cdot T$
4. How many grams of oxygen, O_2 , are there in a 50.0 L gas cylinder at 21°C when the oxygen pressure 15.7 atm ?
 - a. $1.04 \times 10^3 \text{ g O}_2$
 - b. $1.78 \times 10^3 \text{ g O}_2$
 - c. $3.24 \times 10^3 \text{ g O}_2$
 - d. $10.4 \times 10^4 \text{ g O}_2$
 - e. $1.39 \times 10^2 \text{ g O}_2$
5. A sample of a gaseous substance at 25°C and 0.862 atm has a density of 2.26 g/L . What is the molecular mass of the substance?
 - a. 154 g/mol
 - b. 77 g/mol
 - c. 130 g/mol
 - d. 127 g/mol
 - e. 127.8 g/mol

6. Gaseous _____ is the process whereby a gas spreads out through another gas to occupy the space uniformly.
7. _____ states the relation between the effusion of a gas to its molecular mass.
8. Write the van der Waals equation for the pressure correction of an ideal gas.
9. Draw the plot of volume vs. pressure for a sample gas at a temperature?
10. What is partial pressure?